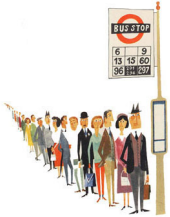
Stacks and Queues

Dr. Anirban Ghosh

School of Computing
University of North Florida

Stacks and Queues

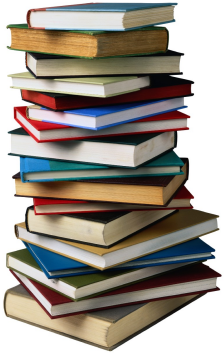


- **Stacks** are **Queues** are two popular easy-to-implement **abstract data types** (ADTs) used in algorithm design
- For an abstract data type, we **describe** the operations that can be executed on the data, but we do not define them
- For instance, the stack ADT can be implemented using a linked-list and an array as well

Stack

Definition

A stack is an ADT where items are inserted (**pushed**) and removed (**popped**) according to the **last-in, first-out** (LIFO) principle



Operations on a stack

Let us denote a stack by S.

1. `S.push(e)`: **adds** element e to the top of S
2. `S.pop()`: **removes** and **returns** the top element from S; returns `null` if S is empty
3. `S.top()`: **returns** the top element from S but does not remove it
4. `S.size()`: **returns** the number of elements stored in S currently
5. `S.isEmpty()`: **returns** `true` if S is empty otherwise returns `false`

push(55)

55

push(3)

$$55, 3$$

`push(69)`

$$55, 3, 69$$

`push(-88)`

$$55, 3, 69, -88$$

push(-17)

$$55, 3, 69, -88, -17$$

`isEmpty()` returns `false`

$$55, 3, 69, -88, -17$$

`size()` returns 5

$$55, 3, 69, -88, -17$$

`pop()` removes and returns -17

$$55, 3, 69, -88$$

`pop()` removes and returns -88

$$55, 3, 69$$

`top()` returns 69

$$55, 3, 69$$

Applications

- Can be used to implement the **FORWARD** and **BACK** buttons in a browser.
- Used by text editors to implement the **UNDO** operation. When you change something, the current state is pushed onto the stack before applying the new change. Hitting UNDO will simply revert the document to the previous state
- Method calls, recursion, and compiler design
- Used as an auxiliary data structure for many cornerstone algorithms

An interface for stacks

```java
public interface StackADT<E> {
   int size();
   boolean isEmpty();
   void push(E e);
   E pop();
   E top();
}
```

To use a stack, one should create a class by implementing this interface

So how to implement a stack using the `StackADT` interface?

- Use an array (stack size must be known in advance, speedy in practice)
  See the class `ArrayStack`
- Use a linked-list (can grow arbitrarily, a bit slower than the array-based stack)
  See the class `LinkedStack`

Irrespective of the implementation, every one of these five stack operations,
`size()`, `isEmpty()`, `push()`, `pop()`, `top()`, takes $O(1)$ time

An application: matching parentheses

- Given an algebraic expression, how to check if the parentheses, braces, and brackets in it are properly **balanced** or not
- Some **balanced** ones
  - `(a + b) + (66 * (s / t) - 12) * {p + (z - [2 + 99.1 / (m - x)])}`
    Sequence: `( ) ( ( ) ) { ( [ ( ) ] ) }`
  - `30 + (40 + (9 * (6 - 10) * (1 - (17 + 19 + 99) - 1) / {5 + (8 - [6 + 2 * (1 + (-99)) - 1] - 4) - 1} - 1) - 1) * 88`
    Sequence: `( ( ( ) ( ( ) ) { ( [ ( ) ] ) } ) )`
- Some **unbalanced** sequences
  - `) ( ( ) ) { ( [ ( ) ] ) }`
  - `( { [ ] ) }`
  - `(`

How to solve it?

- Recall the three pairs we are interested in:

  `( )`, `{ }`, `[ ]`

- Here is a stack-based algorithm for this problem:
  1. Declare a `Character` stack `S`
  2. Scan the input algebraic expression from left to right
  3. If a left symbol `(`, `{`, or `[` is encountered, **push** it onto `S`
  4. If a right symbol `)`, `}`, or `]` is encountered, check the top element in `S`; if these two elements form a matching pair, then pop from `S` otherwise report **INVALID**
  5. At the end, when the string is fully scanned, if `S` is empty, then report **VALID**, else **INVALID**

Sample run

Current symbol scanned	Stack (right is the stack top)	Action taken
( ) (( )) { ([( )]) }	(	push
( ) (( )) { ([( )]) }		pop
( ) (( )) { ([( )]) }	(	push
( ) (( )) { ([( )]) }	((	push
( ) (( )) { ([( )]) }	(	pop
( ) (( )) { ([( )]) }		pop
( ) (( )) { ([( )]) }	{	push
( ) (( )) { ([( )]) }	{ (	push
( ) (( )) { ([( )]) }	{ ([	push
( ) (( )) { ([( )]) }	{ ([(	push
( ) (( )) { ([( )]) }	{ ([	pop
( ) (( )) { ([( )]) }	{ (	pop
( ) (( )) { ([( )]) }	{	pop
( ) (( )) { ([( )]) }		pop

Result: **VALID**

Another sample run

Current symbol scanned	Stack (right is the stack top)	Action taken
({ [ ]) }	(	push
({ [ ]) }	({	push
({ [ ]) }	({ [	push
({ [ ]) }	({	pop
({ [ ]) }	({	**mismatch**

Result: **INVALID**

Code

```java
import java.util.Scanner;
public class ExpresssionChecker {
   public static void main(String[] args) {
      Scanner input = new Scanner(System.in);
      String expression = input.nextLine();
      LinkedStack<Character> S = new LinkedStack<>();
      int pos;

      for(pos = 0; pos < expression.length(); pos++) {
         char current = expression.charAt(pos);

         if( current == '(' || current == '{' || current == '['        S.push(current);
         else if( current == ')' && !S.isEmpty() && S.top() == '(' )   S.pop();
         else if( current == ')' && !S.isEmpty() && S.top() != '(' )   break;
         else if( current == '}' && !S.isEmpty() && S.top() == '{' )   S.pop();
         else if( current == '}' && !S.isEmpty() && S.top() != '{' )   break;
         else if( current == ']' && !S.isEmpty() && S.top() == '[' )   S.pop();
         else if( current == ']' && !S.isEmpty() && S.top() != '[' )   break;
      }

      if( S.isEmpty() && pos == expression.length() )   System.out.println("VALID");
      else System.out.println("INVALID");

      input.close();
   }
}
```

Food for thought

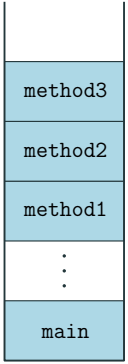
```
<html>
   <head>
      <title>Data Structure</title>
   </head>

   <body>
      <h1>
         <p>Stacks are fun...</p>
      </h1>

      <p>Queues are also fun...</p>
   </body>
</html>
```

Write a program that can check if an HTML file is correctly formatted. This means every opening tag must have its closing tag at an appropriate location in the file.

Method calls and use of stack

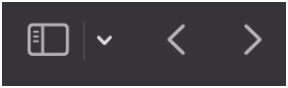
- Ever wondered how the control returns to the **caller** method after the **callee** method is done with its execution?

```
public char method1() {
   // ...
   method2();
   counter++; // control returns here
   i = j + 10;
   //...
}
public double method2() {
   // ...
    s = p + s;
    method3();
    arr[q] = arr[q] / (10 * m); // control returns here
   // ...
}
public void method3() {
   // ...
    s = p + s;
   // ...
   return;
}
```



- A stack of **activation record**s is used by the system; push a record when a method starts execution; pop its record when it is done

Browser tabs: an application of stacks



- For every tab, we get to see the *go forward* and *go backward* buttons
- How to implement this button
- Use 2 **stacks**: one for the *go backward* button and one for the *go forward* button

Code

```java
public class BrowserTab {
   String currentPage = "";
   private final LinkedStack<String> backPages = new LinkedStack<>(), forwardPages = new LinkedStack<>();
   public static String getTimeStamp(){ return "[" + new Date() + "] "; }

   public BrowserTab() { System.out.println(getTimeStamp() + "New tab opened."); }

   public void typeAndGoNewSite(String newPage) {
      if( !currentPage.isEmpty() ) // some page is being viewed currently
      backPages.push(currentPage); // send the current page to the previousPages DLL

      currentPage = newPage;
      System.out.println(getTimeStamp() + "Currently viewing: " + currentlyViewing());
   }
   // contd. on the next slide
}
```

Code

```java
public class BrowserTab {
   public void clickOnGoBackButton() {
      if( backPages.isEmpty() ) {
         System.out.println(getTimeStamp() + "Back button is greyed out (unavailable).");
         return;
      }
      else {
         System.out.print(getTimeStamp() + "Back button is clicked. ");
         forwardPages.push(currentPage);
         currentPage = backPages.top();
         backPages.pop();
      }
      System.out.println("Currently viewing: " + currentlyViewing());
   }

   public void clickOnGoForward() {
      if( forwardPages.isEmpty() ) {
         System.out.println(getTimeStamp() + "Forward button is greyed out (unavailable).");
         return;
      }
      else {
         backPages.push(currentPage);
         currentPage = forwardPages.pop();
      }
      System.out.println(getTimeStamp() + "Forward button is clicked. Currently viewing: " + currentlyViewing());
   }    // contd. on the next slide
}
```

Code

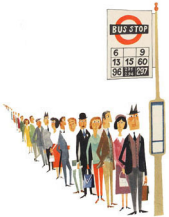
```java
public class BrowserTab {

   public String currentlyViewing() { return currentPage; }

   public String toString() {
      StringBuilder prettyString = new StringBuilder();

      for (String site : previousPages)
         prettyString.append(site).append(" ");

      prettyString.append(" ***").append(currentPage).append("*** ");

      for (String site : nextPages)
         prettyString.append(site).append(" ");

      return prettyString.toString();
   }
}
```

For an usage refer to the class **TestBrowserTab**

Queue

Definition

A queue is an ADT where items are inserted (**enequed**) and removed (**dequeued**) according to the **first-in, first-out** (FIFO) principle



Operations on a queue

Let us denote a queue by `Q`.

1. `Q.enqueue(e)`: **adds** an element `e` to the back of `Q`
2. `Q.dequeue()`: **removes** and **returns** the first element from `Q`; returns `null` if `Q` is empty
3. `Q.first()`: **returns** the first element of `Q`, without removing it; returns `null` is `Q` is empty
4. `Q.size()`: **returns** the number of elements stored in `Q` currently
5. `Q.isEmpty()`: **returns** `true` if `Q` is empty, otherwise returns `false`

enqueue(55)

55

enqueue(3)

$$55, 3$$

enqueue(69)

$$55, 3, 69$$

enqueue(-88)

$$55, 3, 69, -88$$

enqueue(-17)

$$55, 3, 69, -88, -17$$

`isEmpty()` returns `false`

$$55, 3, 69, -88, -17$$

`size()` returns 5

$$55, 3, 69, -88, -17$$

`dequeue()` removes and returns 55

$$3, 69, -88, -17$$

`dequeue()` removes and returns 3

$$69, -88, -17$$

`first()` returns 69

$$69, -88, -17$$

Applications

- Simulation of real-world queues (airlines, ticket counter, etc.)
- Graph algorithms
- Resource sharing in multi-user systems
- Operating systems
- Computer networks

...

An interface for queues

```java
public interface QueueADT<E> {
   void enqueue(E e);
   E dequeue();
   int size();
   boolean isEmpty();
   E first();
}
```

To use a queue, one should create a class by implementing this interface

Implementing using array

0	1	2	3	4	5	6					
Z	N	Z	T	P	A	Y	• • •	S	C	empty	empty

front back

After 3 dequeues and 2 enqueues the situation is: we have space but we cannot enqueue anymore!

0	1	2	3	4	5	6					
empty	empty	empty	T	P	A	Y	• • •	S	C	K	X

front

What to do then?

0	1	2	3	4	5	6				
empty	empty	empty	T	P	A	Y	• • •	S	C	K	X
			front							

- Trivial solution: shift the content to the left and add new stuff
- Downside? This is very expensive! Takes $O(n)$ time
- We need to implement `enqueue` faster than this
- Solution: wrap around

Using the mod operator for wrapping around

0	1	2	3	4
A	B	C	D	E

$$i = 0$$

Next index: $(i + 1)$ mod $5 = 1$ mod $5 = 1$

Using the mod operator for wrapping around

0	1	2	3	4
A	B	C	D	E

$$i = 1$$

Next index: $(i + 1)$ mod $5 = 2$ mod $5 = 2$

Using the mod operator for wrapping around

0	1	2	3	4
A	B	C	D	E

$$i = 2$$

Next index: $(i + 1)$ mod $5 = 3$ mod $5 = 3$

Using the mod operator for wrapping around

0	1	2	3	4
A	B	C	D	E

$$i = 3$$

Next index: $(i + 1)$ mod $5 = 4$ mod $5 = 4$

Using the mod operator for wrapping around

0	1	2	3	4
A	B	C	D	E

$$i = 4$$

Next index: $(i + 1)$ mod $5 = 5$ mod $5 = 0$

Using the mod operator for wrapping around

0	1	2	3	4
A	B	C	D	E

$$i = 0$$

Next index: $(i + 1)$ mod $5 = 1$ mod $5 = 1$

Using this wrap-around approach for array-based queues

0	1	2	3	4	5	6					
Z	N	Z	T	P	A	Y	• • •	S	C	empty	empty

front back

0	1	2	3	4	5	6					
empty	empty	empty	T	P	A	Y	• • •	S	C	K	X

front

0	1	2	3	4	5	6					
empty	empty	empty	T	P	A	Y	• • •	S	C	K	X

back front

Using this wrap-around approach for array-based queues

0	1	2	3	4	5	6					
empty	empty	empty	T	P	A	Y	• • •	S	C	K	X
back			front								

After `Q.enqueue('A');`, we get

0	1	2	3	4	5	6					
A	empty	empty	T	P	A	Y	• • •	S	C	K	X
	back		front								

So how to implement a queue using the `QueueADT` interface?

- Use an array (queue size must be known in advance, speedy in practice)
  See the class `ArrayQueue`
- Use a linked-list (can grow arbitrarily, a bit slower than the array-based queue)
  See the class `LinkedQueue`

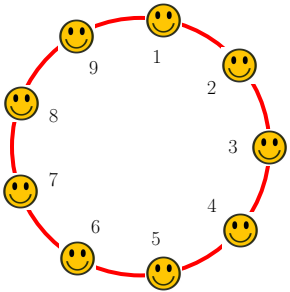
Irrespective of the implementation, every one of these five queue operations,
`size()`, `isEmpty()`, `enqueue()`, `dequeue()`, `first()`, takes $O(1)$ time

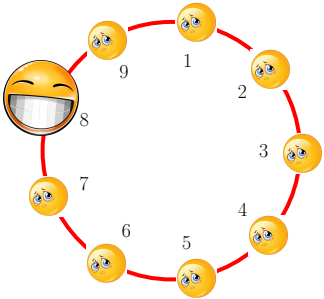
An application: the Josephus problem

The problem

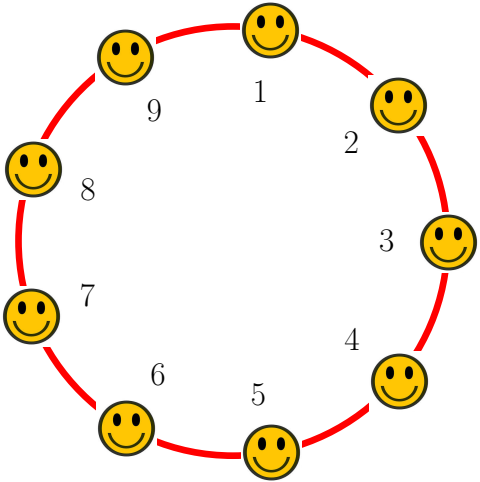
A group of n people agree to play the following fun game. They arrange themselves on a circle (at positions numbered from 1 to n) and proceed around the circle clockwise, eliminating every mth person until only one person is left.

The last remaining person wins the game. How to figure out the winning position?

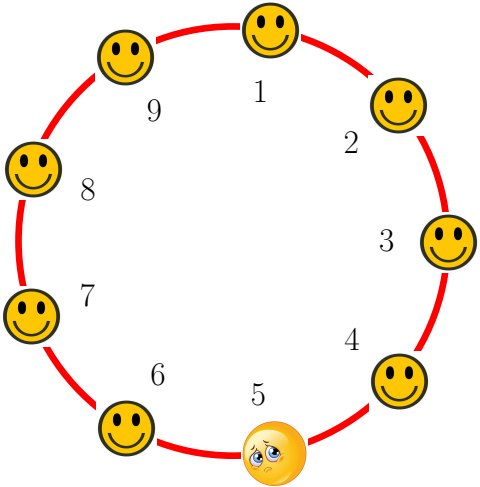




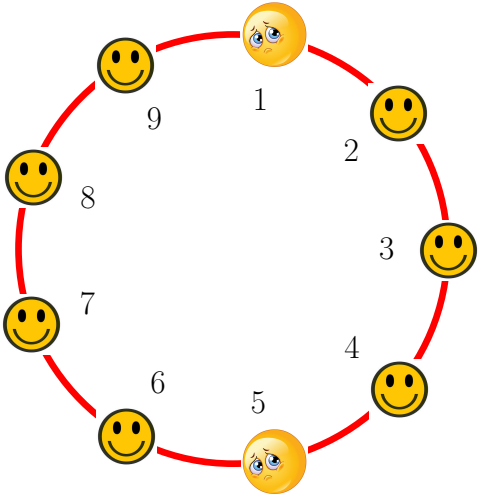
Example: $n = 9, m = 5$



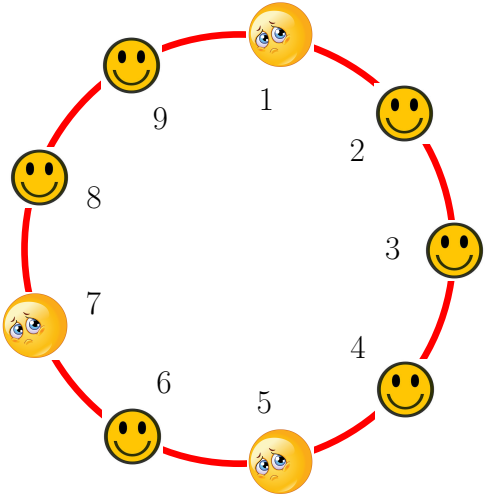
Example: $n = 9, m = 5$



Example: $n = 9, m = 5$



Example: $n = 9, m = 5$



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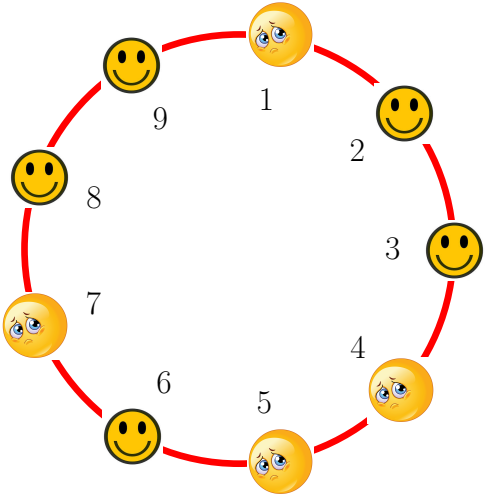
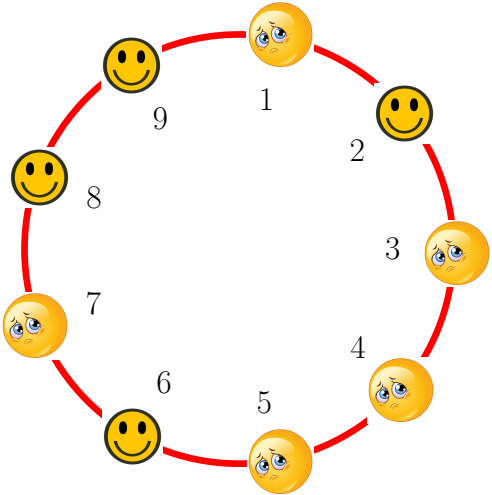
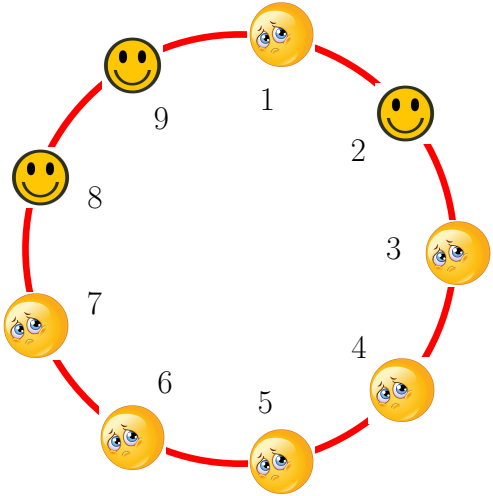


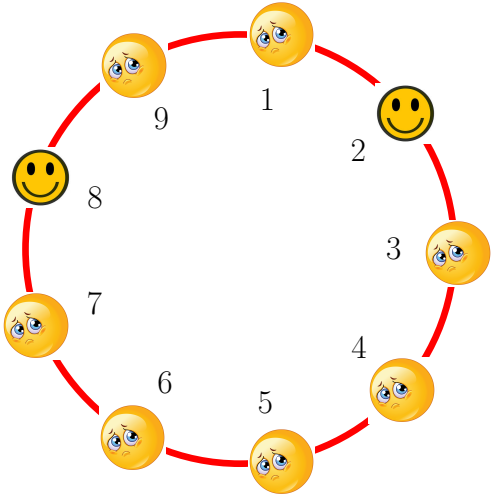
Example: $n = 9, m = 5$



Example: $n = 9, m = 5$



Example: $n = 9, m = 5$



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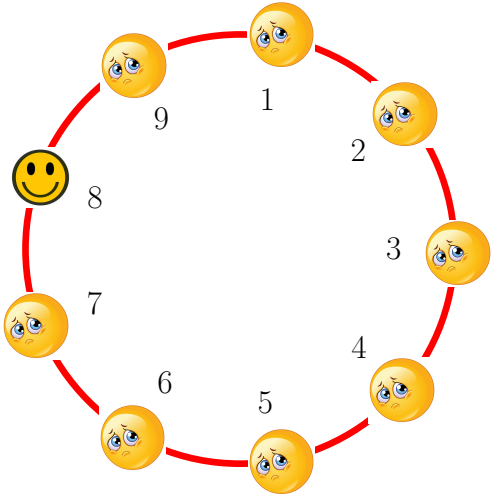
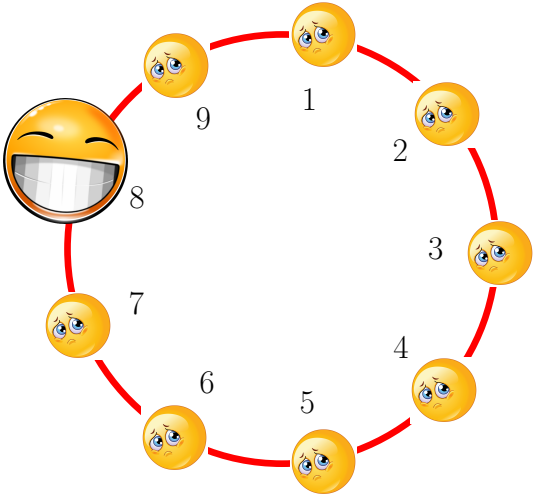


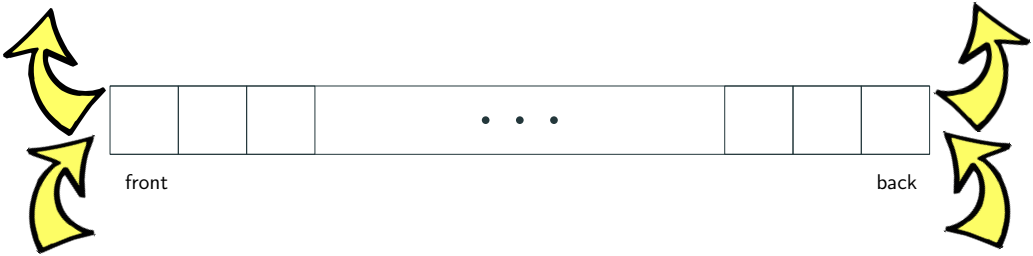
Example: $n = 9, m = 5$



Code

```java
import java.util.Scanner;

public class JosephusSolver {
   public static void main(String[] args) {
      int n = 9, m = 5;

      LinkedQueue<Integer> Q = new LinkedQueue<>();

      for (int i = 1; i <= n; i++) // populate the queue with the integers 1,2,...,n
         Q.enqueue(i);

      while ( (Q.size() > 1) ) {
         for (int i = 0; i < m-1; i++) // send the first m-1 elements to the back of the queue
            Q.enqueue(Q.dequeue());

         System.out.println("Eliminating player: " + Q.dequeue() + " ");
      }

      System.out.print("\nWinning position: " + Q.first()); // the sole integer in the queue is the winner
   }
}
```

Double ended queues: Deque (pronounced as DECK)



Supports insertion and removal at both the front and the back

Deque can be used as a stack or a queue

Deque ADT

```java
public interface DequeADT<E> {
   int size();
   boolean isEmpty();
   E first();
   E last();
   void addFirst(E e);
   void addLast(E e);
   E removeFirst();
   E removeLast();
}
```

An easy way to implement a deque

Doubly linked-list (using a singly linked-list will make the `removeLast` operation run in $O(n)$ time)

☞ Refer to the class `LinkedDeque` for an implementation

Time complexity

Just like stacks and queues, all operations on deques run in $O(1)$ time

Suggested exercise

Create an array-based iterable deque class

Reading

Stacks and queues from **Chapter 5**

https://opendsa-server.cs.vt.edu/OpenDSA/Books/CS3/html/index.html